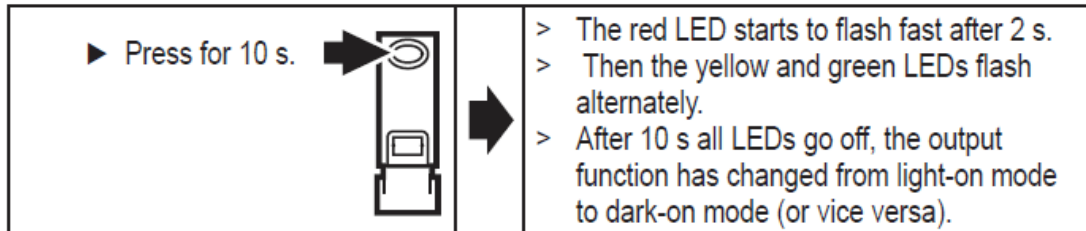


Switching Between Light-On and Dark-On Modes

The OJ5130 and OJ5131 are typically used in a light-on (light-operate) mode. In order to switch between light-on and dark-on (dark-operate) modes, use the following procedure:



Interpreting LED Status for the #OJ5130 / #OJ5131

LED green is lit	Unit is ready for operation.
LED yellow is lit	Output is switched.
LED red is lit	Error in object detection, e.g. maladjustment, soiling of the lenses.
LEDs yellow + red	Flash alternately, 2 Hz: output short-circuited. Flash alternately, 1 Hz: internal malfunction (output is not switched).

Sensor Debounce Setup

With the sensor working properly to consistently “see” the product, the last step in commissioning the sensor is to adjust the “Debounce” timers within the system PLC. A Debounce time is the time that the sensor must be continuously made or un-made before its signal is used to do something. This prevents momentary flickering of the sensor from triggering an action. These timers are accessible through the Engineering Screens on the HMI screen. For each sensor, there is a timer value for both the “Covered” and “Uncovered” debounce value. A debounce timer value is typically anywhere from 250 msec (1/4 second) to 2000 msec (2.0 seconds). A reference engineering screen is shown in Figure PH006-12 below.

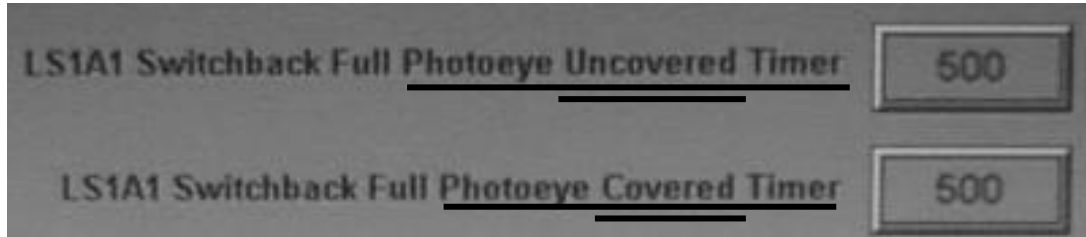


Figure PH006-12 – Engineering screen showing Debounce values(milliseconds)

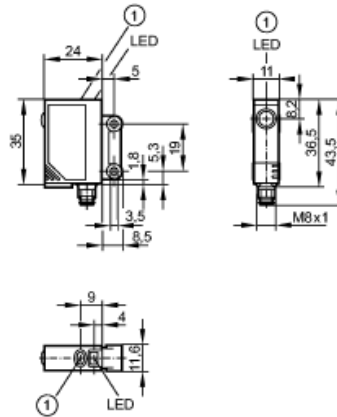
Specifications for the #OJ5130

efector200



OJ5130

Photoelectric sensors



1: Pushbutton without function



Made in Germany

Product characteristics

Through-beam sensor

Rectangular, plastics

Quick disconnect

Side sensing

Transmitter

Sensing range 10m

Electrical data

Electrical design

DC

Operating voltage [V]

10...30 DC

Current consumption [mA]

< 19

Type of light

red light 660 nm

Protection class

III

Reverse polarity protection

yes

Outputs

Overload protection

yes

Monitoring range

Sensing range [m]

10

Light spot diameter [mm]

1000 (at maximum range)

Environment

Ambient temperature [°C]

-25...60

Protection

IP 67

Tests / approvals

EMC

EN 60947-5-2

MTTF

[Years]

1764

Mechanical data

Housing materials

housing: ABS; LED window: SEPS; pushbutton: SEPS

Lens material

glass

Weight

[kg]

0.044

Specifications for the #OJ5130 (cont.)

efector200



OJ5130

Photoelectric sensors

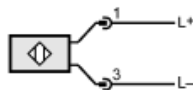
Displays / operating elements

Power	LED	green
-------	-----	-------

Electrical connection

Connection	M8 connector
------------	--------------

Wiring



Accessories

Accessories (included)	2 mounting screws; 2 spring washers; 2 nuts
------------------------	---

Remarks

Remarks	cULus - Class 2 source required
---------	---------------------------------

Pack quantity	[piece]	1
---------------	---------	---

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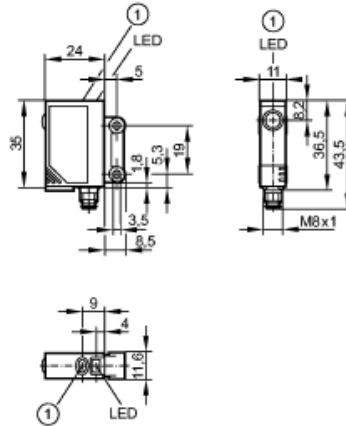
Specifications for the #OJ5131

efector200



OJ5131

Photoelectric sensors



1: pushbutton



Made in Germany

Product characteristics

Through-beam sensor
Rectangular, plastics
Quick disconnect
Electronic lock
External teach function
Side sensing
Receiver
Sensing range 10m
adjustable

Electrical data

Electrical design	DC PNP
Operating voltage [V]	10...30 DC
Current consumption [mA]	< 12
Protection class	III
Reverse polarity protection	yes

Outputs

Output function	light-on / dark-on programmable
Voltage drop [V]	< 2.5
Current rating [mA]	200
Short-circuit protection	Yes (non-latching)
Overload protection	yes
Switching frequency [Hz]	1200

Monitoring range

Sensing range [m]	10
-------------------	----

Excess gain graphs

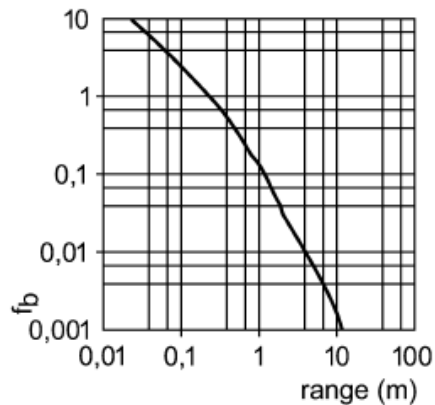
Specifications for the #OJ5131 (cont.)

efector200



Photoelectric sensors

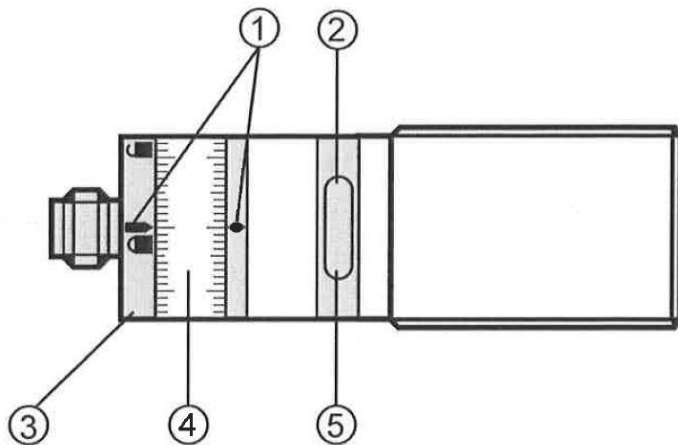
OJ5131



Environment		
Ambient temperature	[°C]	-25...60
Protection		IP 67
Tests / approvals		
EMC		EN 60947-5-2
MTTF	[Years]	952
Mechanical data		
Housing materials		housing: ABS; LED window: SEPS; pushbutton: SEPS
Lens material		glass
Weight	[kg]	0.043
Displays / operating elements		
Output status indication	LED	yellow
Power	LED	green
Function display	LED	red
Electrical connection		
Connection		M8 connector
Wiring 		
Accessories		
Accessories (included)		2 mounting screws; 2 spring washers; 2 nuts
Remarks		
Remarks		cULus - Class 2 source required
Pack quantity	[piece]	1

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Optical Distance Sensor – efector OID250 (H&C Part #20591952) Controls Setup and Line Commissioning



1: Setting Marks

2: LED Yellow: Set 1 value reached, output=ON

3: Locking Ring

4: Setting Ring (manually adjustable after unlocking)

5: LED Green: Operating voltage OK

Note: *The OID250 Optical Distance Sensor is a Class 1 Laser Product, which is personnel safe, but always use caution when installing and adjusting the sensor.*

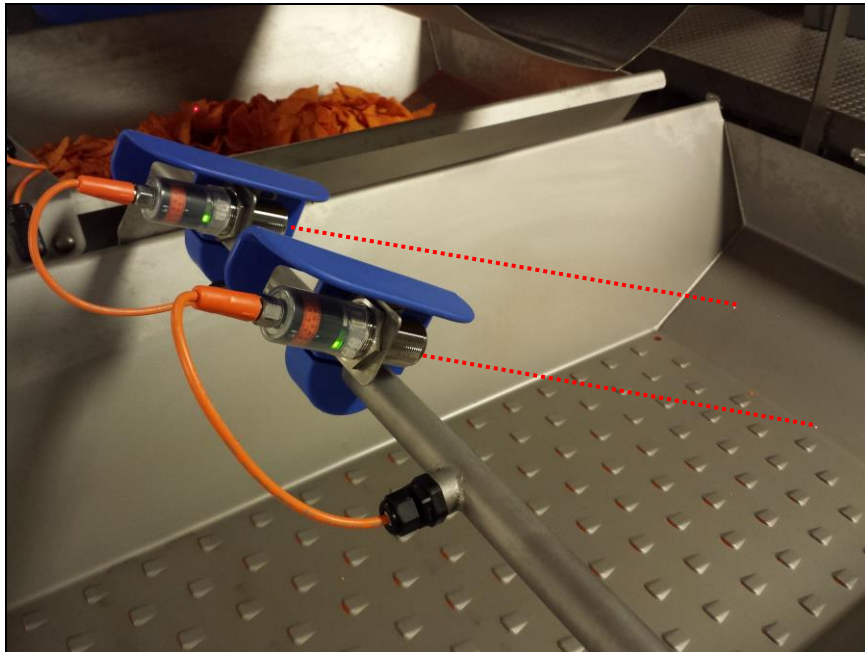
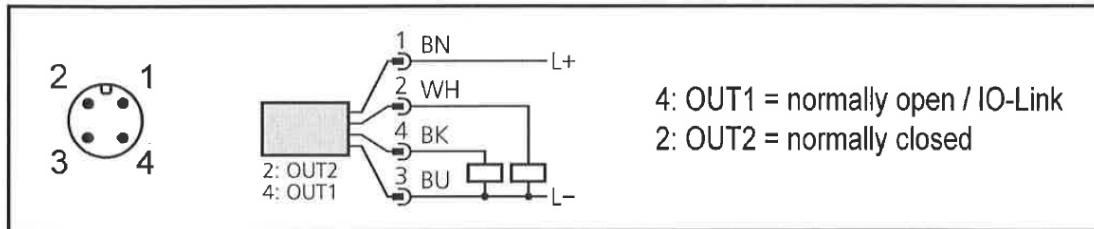


Photo PH006-6 – OID250 Installed

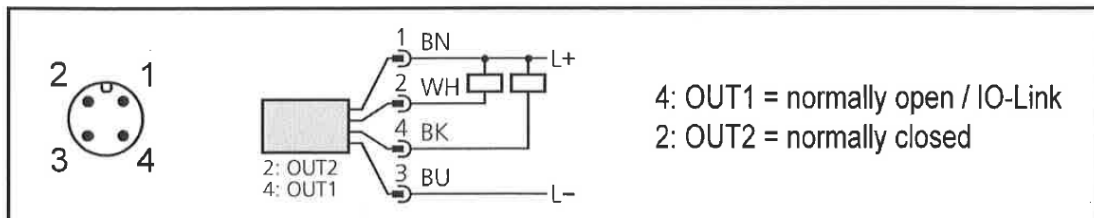
Wiring the OID250

- Disconnect power.
- Connect the unit as follows:

DC PNP



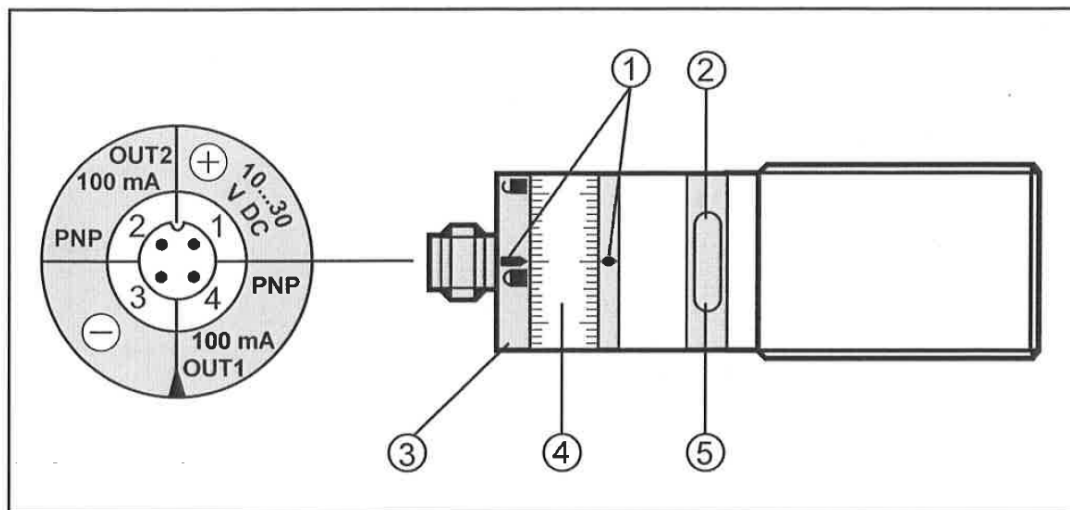
DC NPN



Core colours of ifm sockets:

1 = BN (brown), 2 = WH (white), 3 = BU (blue), 4 = BK (black)

Calibrating the OID250



- 1: Setting marks
 - 2: LED yellow: Set1 value reached, output = ON
 - 3: Locking ring
 - 4: Setting ring (manually adjustable after unlocking)
 - 5: LED green: operating voltage OK
- To achieve the setting accuracy: First position the setting ring to the upper end stop value, then to the requested value.

Locking and Unlocking the OID250

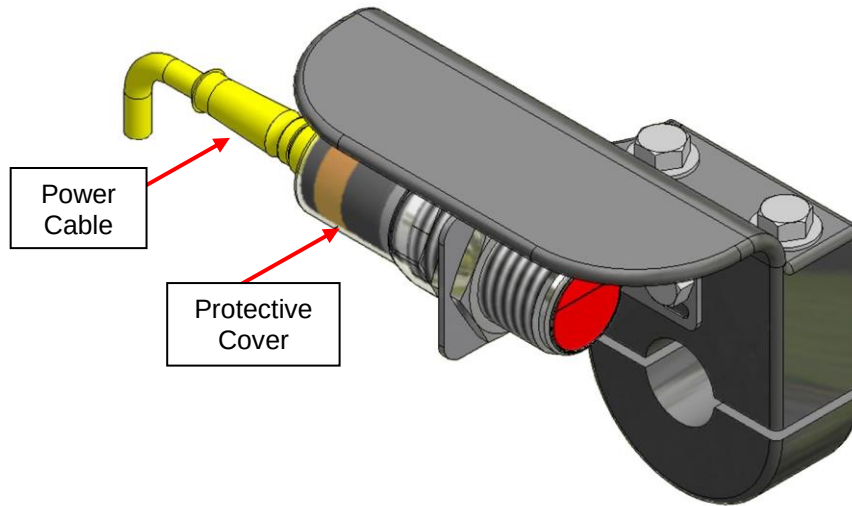


Figure PH006-13 – OID250 Protective Cover

After calibrating the distance, lock the sensor.

To Unlock:

Rotate the Locking Ring to where the setting marks for Unlock  line up with the smaller setting mark (see Figure PH006-14).

To Lock:

Rotate the Locking Ring to where the setting marks for Lock , line up with the smaller setting mark (see Figure PH006-14)

Figure PH006-14 shows the OID250 locked.

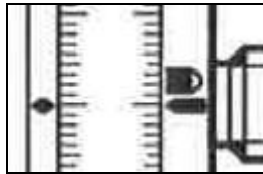


Figure PH006-14 – Example of Setting Marks

Final Steps:

- 1. Remove the power cable from the back of the sensor (see Figure PH006-13).**
- 2. Replace the protective cover.**
- 3. Replace the power cord.**
- 4. Do a final check of the sensor to verify that it is detecting the correct distance.**

Sensor Installed

Photo PH006-7 – OID250 Installed shows a close-up of a pair of OID250s installed.

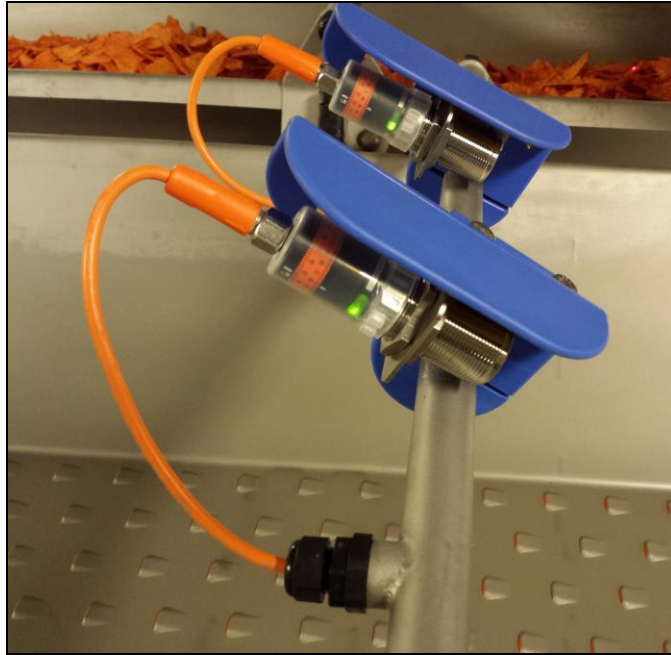


Photo PH006-7 – OID250 Installed (close-up)

Sensor Debounce Setup

With the sensor working properly to consistently “see” the product, the last step in commissioning the sensor is to adjust the “Debounce” timers within the system PLC. A Debounce time is the time that the sensor must be continuously made or un-made before its signal is used to do something. This prevents momentary flickering of the sensor from triggering an action. These timers are accessible through the Engineering Screens on the HMI screen. For each sensor, there is a timer value for both the “Covered” and “Uncovered” debounce value. A debounce timer value is typically anywhere from 250 msec (1/4 second) to 2000 msec (2.0 seconds). A reference engineering screen is shown in Figure PH006-15 below.

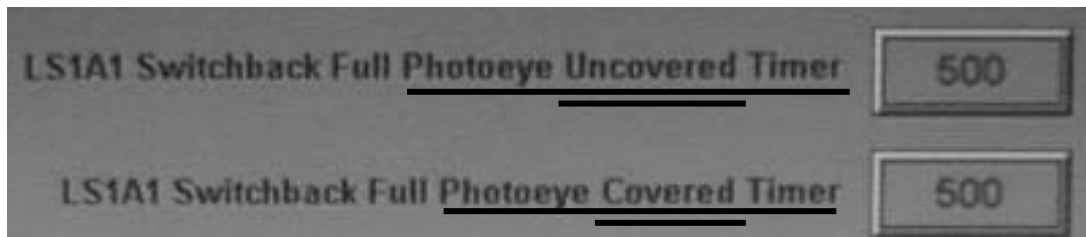


Figure PH006-15 – Engineering screen showing Debounce values (milliseconds)

Specifications for the OID250

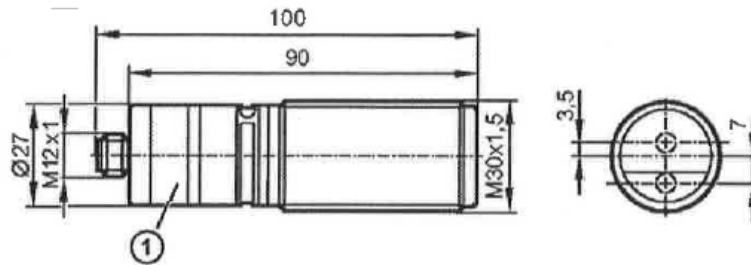
efector200



OID250

OIDLCPKG/US

Photoelectric sensors



1: setting ring



Made in Germany

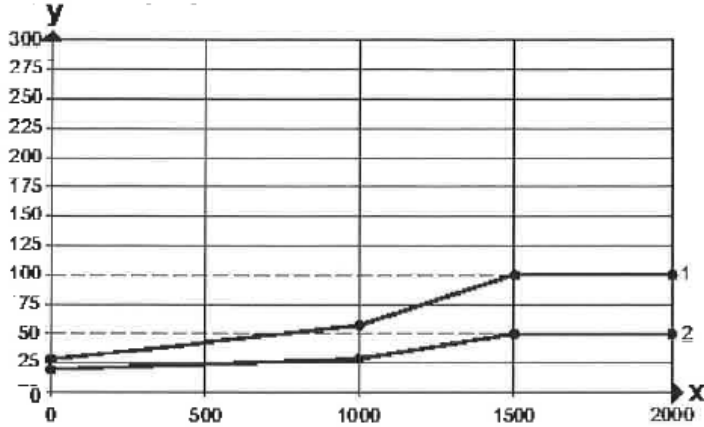
Product characteristics	
Photoelectric distance sensor	
Quick disconnect	
Background suppression	
Visible laser light, LASER CLASS 1	
setting ring	
manually adjustable [cm/inch]	
Measuring range 0.03...2 m	
Background suppression ...20 m	
Electrical data	
Electrical design	DC PNP
Operating voltage [V]	10...30 DC
Current consumption [mA]	75; (24 V)
Life expectancy typ. [h]	50000
Type of light	red light 650 nm
Protection class	III
Reverse polarity protection	yes
Outputs	
Output function	2 switching outputs normally open / closed complementary
Current rating [mA]	2 x 100
Short-circuit protection	Yes (non-latching)
Overload protection	yes
Switching frequency [Hz]	11
Monitoring range	
Background suppression	...20 m
Light spot diameter [mm]	< 5 (Range 2 m)
Hysteresis [%]	< 5; black (6 % remission) at max. range

Specifications for the OID250 (cont.)

Hysteresis graph

x: distance in [mm]

y: Hysteresis in [mm]



1 = object black (6 % remission)
2 = object white (90 % remission)

Measuring / setting range

Measuring range [m]	0.03...2
Sampling rate [Hz]	33

Interfaces

IO-Link Device

Transfer type	COM2 (38.4 kBaud)
IO-Link revision	1.1
SDCI standard	IEC 61131-9 CDV
IO-Link Device ID	373d / 00 01 75h
SIO mode	yes
Process data analogue	-
Process data binary	2
Min. process cycle time [ms]	6.6

Environment

Ambient temperature [°C]	-25...60, at ta < -10 °C warm-up is necessary, laser is off
Protection	IP 65 / IP 67

Tests / approvals

EMC	EN 60947-5-2
-----	--------------

Mechanical data

Housing materials	housing: stainless steel; window: PMMA; LED window: polycarbonate; PBT (Pocan); PC (Makrolon); FPM (Viton)
Weight [kg]	0.206

Displays / operating elements

Display	Switching status	LED yellow
	Power	LED green
Display unit	cm, inch	

Electrical connection

Connection	M12 connector
------------	---------------

Specifications for the OID250 (cont.)

Wiring



4: OUT / IO-Link

Accessories	
Accessories (Included)	2 lock nuts (stainless steel); M30
Remarks	
Remarks	Laser light CLASS 1 LASER PRODUCT IEC 60825-1:2007 complies with 21 CFR 1040.10 except for deviations pursuant to Laser Notice No. 50, dated June 2007 cULus - Class 2 source required
Pack quantity [piece]	1

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TECHNICAL PUBLICATION PH006 – EFECTOR PHOTOEYE SETUP

Heat and Control Product Handling Systems

Revision Record

Revision Level	Date	By	Description
Revision 00	07/20/06	Tom Knodell	----
Revision 01	07/13/12	Michael Petri	<u>Major Revision</u> • Created Table of Contents • Created efector / H&C Part Numbers section • Created sections for OGT302, O5H500, and OJ5130/OJ5131 • Created Revision Record section
Revision 02	03/24/16	Ken Petri & Arlene Mason	• Updated format. • Added OID250 sensor information.

Technical Publication PH009

Yaskawa V1000 *Setup Procedure*

&

Validated Parameter Lists by Application
With Yaskawa V7 Reference Parameters

Rev. 10



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Setup Procedure for Yaskawa V1000

This procedure describes how to set or restore the Heat and Control factory parameter settings for the **Yaskawa V1000** and how to read and copy those parameters to other inverters of the same power and voltage ratings. If following these steps do not correct the problem, call the Heat and Control, Inc. Service Office in Lititz, PA at (717) 519-5240 or (800) 648-5313.

Initial Setup

For initial setup of the V1000, the following tools are required:

- A #2 Phillips screwdriver
- A small flathead screwdriver
- Yaskawa V1000 hand-held operator (part no. 20548664)



Photo PH009-1 – Required Tools

1. Disconnect Power to the FastBack®

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

Disable power provided to the FastBack® system at the source by turning off the power at the power distribution panel.



WARNING

Allow the system a cool down time of 30 seconds to avoid any chance of electrocution.

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located in a separate compartment behind the accessory panel. Do not operate with chassis cover removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

2. Disconnect communication cable from inverter

Disconnect the silver communication cable from the inverter by depressing the locking tab on the connector and pulling straight out.



Photo PH009-2 – Disconnect the Communications Cable

3. Remove inverter front cover

To remove the front cover of the inverter, loosen the retaining screw and pull the cover off.

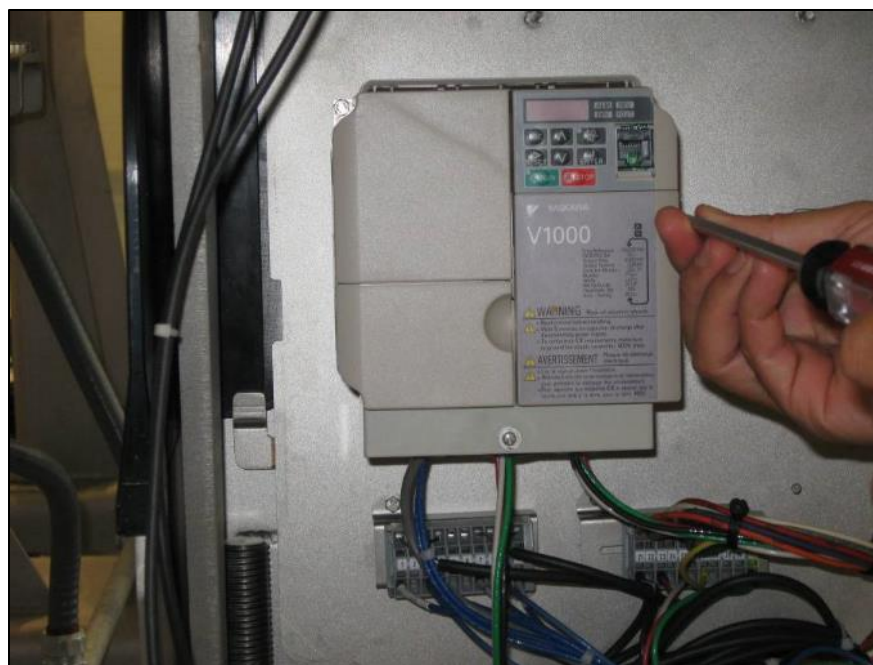


Photo PH009-3 – Loosen Retaining Screw



Photo PH009-4 – Inverter Front Cover Removal

4. Set Main Frequency Reference

For Single-Speed and Multi-Speed Applications:

Set DIP switch S1 to I (right side) and DIP switch S2 to OFF (left side).

For Variable Speed (4-20mA analog input) Applications:

Set the DIP switch S1 to I (right side) and DIP switch S2 to ON (right side).

5. Set Sinking/Sourcing Mode

Using a small flathead screwdriver set DIP Switch S3 to the SINK position (top).

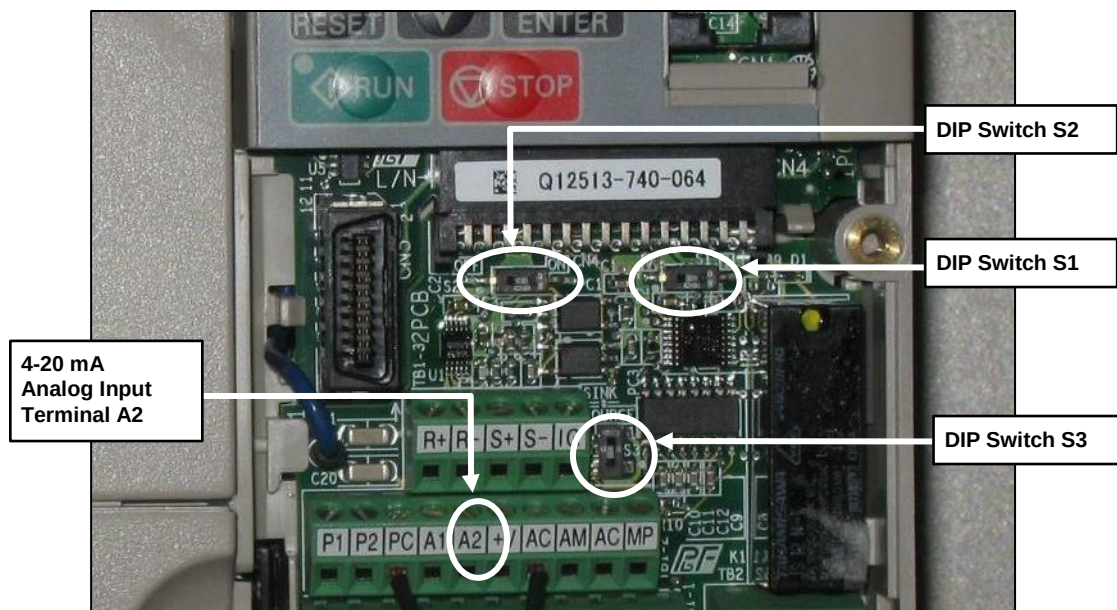


Photo PH009-5 – DIP Switch Locations (4-20mA analog input setup shown)

6. Reinstall the inverter front cover and prepare for operation.

To reinstall the front cover of the inverter, tighten the retaining screw.

Re-connect the silver communication cable to the inverter by depressing the locking tab on the connector, pushing straight in, and releasing the tab.

Using the Handheld LCD Operator

The LCD Operator allows the user to:

- Read, modify, or recover Yaskawa V1000 factory default inverter parameters.
- Copy inverter parameter settings to another Yaskawa V1000 inverter (of the same voltage and power rating).
- Reset drive parameters to Yaskawa factory settings.
- Operate the FastBack® in local mode.
- Monitor important inverter functions and view drive status.
- Customize certain parameters, i.e.: preset speeds.
- Perform field parameter checks and comparisons to this document.
- Store Heat and Control parameter sets.
- Recover Heat and Control parameter sets.

The LCD Operator can be connected to the inverter by one of two ways:

- By plugging the RJ45 connector into the communication port on the front of the inverter.
- By plugging the RJ45 connector into the external communication port on the FastBack®.

The latter is the preferred method.

1. Supply Power to the FastBack®

Connect power to the FastBack® by rotating the emergency disconnect switch to the ON position.

Enable power provided to the FastBack® system at the source by turning on the power at the power distribution panel.



WARNING

Ensure the correct power voltage is supplied to the FastBack®.

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located in a separate compartment behind the accessory panel. Do not operate with housing cover removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

2. Connecting the Operator

Locate the communication port behind the FastBack® Accessory Panel on a Gen2 260E (see Photo PH009-6).



Photo PH009-6 – External Communication Ports (Rev Gate/FastBack® 260E)

Remove the dust cap by turning it counterclockwise.

Plug the cable of the handheld operator into the RJ45 port by pushing the connector straight in. The locking tab should click into position when the connector is fully seated (see Photo PH009-7).



Photo PH009-7 – Communication Cable Plugged Into Port on a Gen2 260E

TECHNICAL PUBLICATION PH009 – YASKAWA V1000 SETUP PROCEDURE

Heat and Control Product Handling Systems

When the operator is plugged in, an initial display showing the Yaskawa logo will appear followed by a second display which provides drive information.

The Frequency Reference screen appears after the initial displays (see Photo PH009-8).

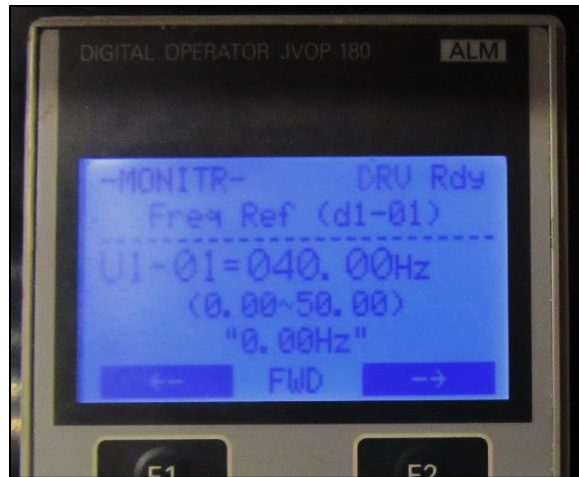


Photo PH009-8 – V1000 Operator Display Showing Frequency Reference Screen

NOTE: When the Handheld LCD Operator is installed and connected to the V1000 it will cause the drive's built-in LED Operator to display a series of dots as shown in Photo PH009-9. This is normal.

Additionally, all of the keys except the STOP key on the inverter's built-in LED Operator will not function when the Handheld LCD Operator is connected.

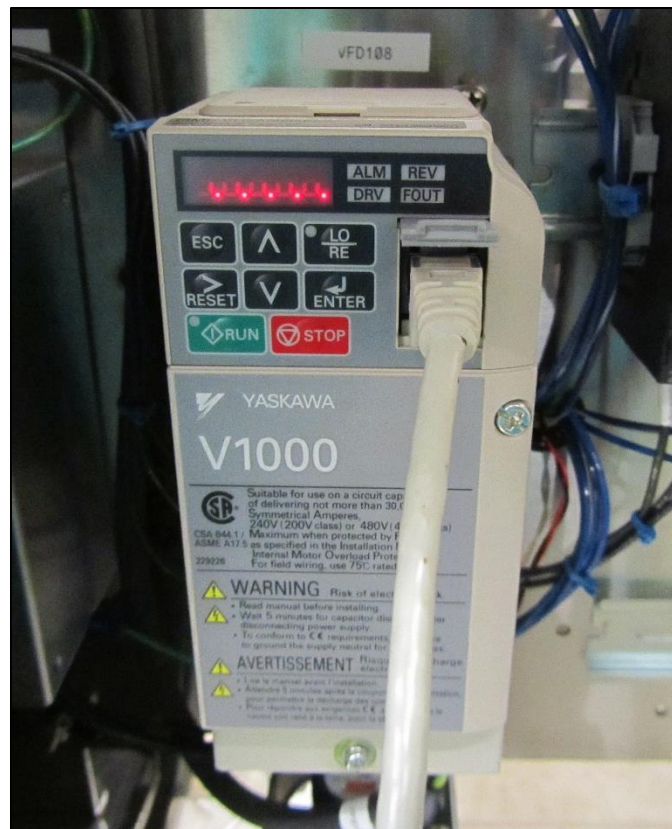


Photo PH009-9 – V1000 Display with Handheld LCD Operator Connected

3. Restore Yaskawa Factory Default Values

Restore the **Yaskawa** factory default values, because VFDs purchased in different parts of the world may have parameter sets defined for that region. This step guarantees that the parameter set in the VFD is a known baseline.

This is accomplished in two steps: 1) Erase the current parameter data in register 1110 and 2) Load the factory defaults from register 2220.

- a. From the Frequency Reference screen, press ▲ or ▼ to navigate to the Programming Mode and press ↵.
- a. In Programming Mode, use ►, ▲ and ▼ to navigate to o2-03 “User (Parameter) Default Sel” and press ↵.
- b. Use ▲ and ▼ to select a value of 2 “Clear User Init (Values)” and press ↵. Display message will read “Entry Accepted”. This will erase previously saved user parameter values.
- c. Now, use ►, ▲ and ▼ to navigate to A1-03 “Initialize Parameter (Values)” and press ↵.
- d. Use ▲ and ▼ to select 2220 “2-Wire Init”.
- e. Press ↵ to store the new settings. The display message will read “Entry Accepted” and returns to a value of 0.
- f. Press [Esc] to return to the Frequency Reference screen.

4. Setting Parameters

Perform the following procedure to set the parameters in the V1000. Table PH009-2 through **Table PH009-9** list validated parameters for various applications. Select the appropriate list for your application.

Note: If your application is not represented here, please call Heat and Control for assistance.

These tables list only the parameters to be changed from the baseline values established in Step 3.

- a. From the Frequency Reference screen, press ▲ or ▼ to navigate to the Programming Mode and press ↵.
- b. Use ►, ▲ and ▼ to navigate to the desired parameter number. The parameter name, setting range and current value will be displayed.
- c. Press ↵ to edit the parameter.
- d. Using ►, ▲ and ▼ to enter the desired value.
- e. Press ↵ to save the new setting. Display message will read “Entry Accepted”.
- f. When complete, press [Esc] or use ►, ▲ and ▼ to navigate to the next parameter on the list.

5. Save VFD Parameters in a User Defined Register Using the Handheld

This function saves the current parameter set in a special memory register in the VFD. If the parameters are altered after this operation, they can be recovered from this register. Normally, the register already contains the parameter set that was loaded at Heat and Control factories. Nevertheless, it is good practice to perform this operation on every VFD, especially if some parameter customization was necessary, as with Upgrade™ and SwitchBack™ accumulation conveyor speeds.

Follow these steps to save VFD settings.

- a. Verify that all parameters match the desired settings.
- b. Set parameter o2-03, “User (Parameter) Default Sel”, to 1 and press . The current parameter values will be saved, “Entry Accepted” will momentarily display, and the displayed value will return to 0.

6. Read Parameter Values

If more than one inverter is to be initialized, read and copy the values from the programmed inverter to the hand-held operator using the steps below:

- a. From the Frequency Reference screen, press  or  to navigate to the Programming Mode and press .
- b. Use ,  and  to select o3-02 “Read Allowable”.
- c. Press .
- d. Use  and  to select 1 “Enabled”.
- e. Press  to enter and store the new setting. Display message will read “Entry Accepted”.
- f. Use ,  and  to select o3-01 “Copy Function Sel”.
- g. Press .
- h. Use  and  to select 1 “INV → OP READ”.
- i. Press . The hand-held operator will read parameter settings from the inverter. Display message will read “READ, INV → OP READING”.
- j. When complete, display message will read “End, READ COMPLETE.”
- k. The Copy Function Selection display will reappear and the displayed value will return to 0.
- l. Press [Esc] to return to the Frequency Reference screen.

7. Copy Parameter Values

Now that the required parameter values are saved to the hand-held operator, copy the values from the hand-held operator to another Yaskawa inverter of the same voltage and power, using the steps below:

- a. From the Frequency Reference screen, press  or  to navigate to the Programming Mode and press .
- b. Use ,  and  to navigate to o3-02 “Read Allowable” and press .

- c. Use ▲ and ▼ to select to 1 “ENABLED”.
- d. Press ↵ to store the new setting. Display message will read “Entry Accepted”.
- e. Use ►, ▲ and ▼ to navigate to o3-01 “Copy Function Sel” and press ↵.
- f. Use ▲ and ▼ to select to 2 “OP → INV WRITE”.
- g. Press ↵. The hand-held operator will copy parameter settings to the inverter. The display message will read “COPY, OP → INV COPYING”.
- h. When complete, display message will read “End, COPY COMPLETE”.
- i. The Copy Function Selection display will reappear and the displayed value returns to 0.
- j. Press [Esc] to return to the Frequency Reference screen.
- k. Save the parameters in the current VFD by performing Step 5.

8. Verify Parameter Values

Once parameters are copied from the hand-held operator to the VFD, they can be verified to ensure the correct settings have been captured. Use the following steps to verify the parameter values.

- a. From the Frequency Reference screen, press ▲ or ▼ to navigate to the Programming Mode and press ↵.
- b. Use ►, ▲ and ▼ to navigate to o3-01 “Copy Function Sel” and press ↵.
- c. Use ▲ and ▼ to select to 3 “OP → INV VERIFY”.
- d. Press ↵. The hand-held operator will verify parameter settings in the inverter. The display message will read “VFY, Data Verifying”.
- e. When complete, display message will read “End, VERIFY COMPLETE”.
- f. The Copy Function Selection display will reappear and the displayed value returns to 0.
- g. Press [Esc] to return to the Frequency Reference screen.

9. Recover the Heat and Control User Defined Parameter Set (A1-03)

If previously saved, the original Heat and Control factory parameter sets or field modified parameter sets can be recovered by setting parameter A1-03, “Initialize Parameters” to a value of 1110. When you press ↵, all parameters (except A1-02, C6-01, E1-03, o2-04, and L8-35) will reset back to their Heat and Control default values, and the value of A1-03 will automatically return to 0. If there is no previously saved parameter set, 1110 is not an option.

Chapter 5 of the Yaskawa V1000 Quick Start Guide lists possible error codes for the READ, COPY and other functions along with their meanings. The Yaskawa V1000 Series Drive Manual (available online or on a CD) has more information on these functions.

Troubleshooting

The section covers some common troubleshooting steps.

1. Verify Parameters (Hand-held and VFD Operator)

The Modified Constants/Parameters function is the easiest way to check what parameters have been altered. It displays all parameters that have been modified from Yaskawa's factory default parameter set for 2-Wire operation and displays the current setting. If using the V1000 Handheld Operator (20548664), the parameter name, default setting, and the setting range will also be displayed. Scroll through the parameter list using the ▲ and ▼ keys. While in this mode, modified parameters may be further modified by pressing ↵ to access that parameter value. After modifying the value, press ↵ to store the new value. If the value is modified to its (default) value, it will disappear from this list.

2. Fault History Display

The fault history display will assist in troubleshooting fault issues. The last ten (10) faults are displayed here. Use the following steps to access the Fault History Display.

- From the Frequency Reference screen, press ▲ or ▼ to navigate to the Monitor Menu and press ↵.
- Use ►, ▲ and ▼ to navigate to U3-01 "Most Recent Fault"
- Use ▲ and ▼ to scroll through the 10 most current faults.
- Press [Esc] to return to the Frequency Reference screen.

Monitor parameters U2-1 through U2-17 give detailed information about the most current and previous faults.

3. Status Monitors

Status monitors display the current status of many critical functions. Use the following steps to access the status monitors.

- From the Frequency Reference screen, use ►, ▲ and ▼ to navigate through the status monitors.

Table PH009-1 Table PH009-1 is a list of the most common monitors to check while troubleshooting.

Register	Name	Function
U1-01	Frequency Reference	Monitors the frequency.
U1-02	Output Frequency	Displays the output frequency.
U1-03	Output Current	Displays the output current.
U1-06	Output Voltage Reference	Displays the output voltage.
U1-07	DC Bus Voltage	Displays the DC bus voltage.

Table PH009-1 – U1 Operation Status Monitors

More information on the Status Monitors can be found in the Yaskawa V1000 Quick Start Guide or the Yaskawa V1000 Series Drive Manual (available online or on a CD).

Additional Information

Wiring Diagram

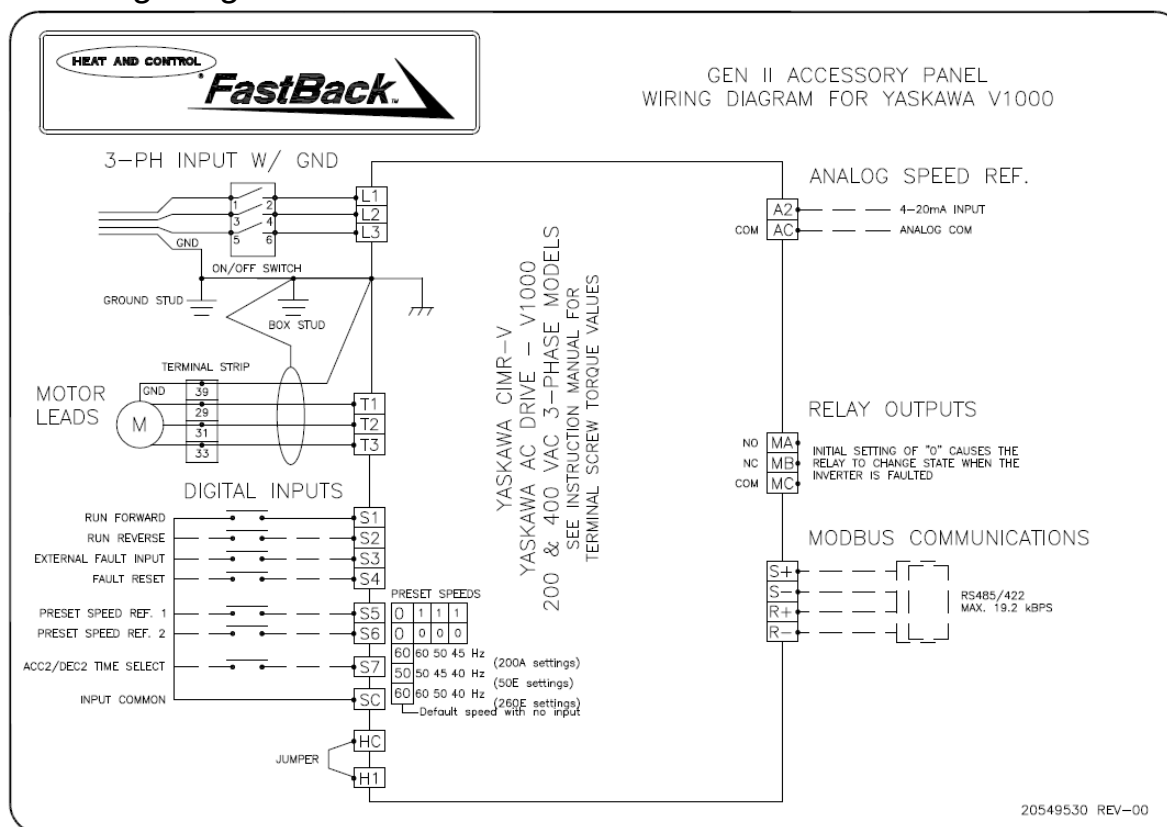


Figure PH009-1 – Typical V1000 Wiring Diagram

These tables list only the parameters to be changed from the baseline values established in Step 3 on page 9.

Yaskawa V1000 Inverter Parameter Values for FastBack® Conveyors

Yaskawa V1000 Inverter Parameter Values for FastBack® Conveyors					
V1000 Parameter Number	V7 Parameter Reference	50E / 90E Values	260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Function Description
<i>The following parameters are required for all applications:</i>					
B1-01	n004	0 (Single/Multi-speed) 1 (Analog Input)			Reference frequency source selection. For single and multi-speed control: Dipswitch S2 = OFF. For analog input: Dipswitch S1 = I (4-20mA) Dipswitch S2 = ON.
B1-04	n006	1			Reverse disabled. (Default: 0)
B1-17	-	1			Run Command at Power Up 1 = Run Command issued, motor operation start
B2-03	-	0.00			Sets DC Injection braking time at start.
B2-04	n090	0.00			Sets DC injection braking time at stop
B8-01	N/A	0 (Default)			Energy Saving Control Selection 0 = Disabled / 1 = Enabled This feature not applicable to the FastBack mechanism – Do Not Enable
C1-01	n019	0.6	0.6 - *	1.0	Sets acceleration time #1 in seconds
C1-02	n020	1.0	1.0 - *	1.5	Sets deceleration time #1 in seconds
C2-03		0.2 sec (Default)		1.0 sec	S-Curve characteristic at decel.
C4-01	N/A	0.00			Torque Compensation Gain (Default: 1.00) CRITICAL SETTING Any value > 0.00 will increase risk of an overvoltage fault
C6-01	N/A	0			Normal/Heavy Duty Selection (Default: 1=Normal Duty)
C6-02	n080	1 (2.0 kHz)			Carrier Frequency Selection (Default depends on O2-04 for 200V and 400V Class)
D1-01	n024	50	60 most applications 55 for Revolution OMS apps	53 for all applications	Sets master speed for single-speed operation (Default: 0 Hz)
E1-04	n011	50	60 (Default)	53	Sets max output freq (Default: 60 Hz)
E1-05	n012	150	130	90	Sets maximum voltage out For 208 / 230 VAC incoming power
		or 300	or 260	or 180	CRITICAL SETTING For 400 / 460 VAC incoming power
E1-06	n013	50	60 (Default)	50	Sets frequency at which voltage output is maximized.

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Yaskawa V1000 Inverter Parameter Values for FastBack® Conveyors						
V1000 Parameter Number	V7 Parameter Reference	50E / 90E Values		260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Function Description
E2-01	n036	1200 rpm				Sets motor rated (nameplate) current.
		1/3 HP 2.55	1/4 HP 2.10	4.40	7.60	For 208 VAC applications
		2.30	1.90	4.00	6.80	For 230 VAC applications
		1.35	1.10	2.30	4.00	For 400 VAC applications
		1.15	0.90	2.00	3.40	For 460 VAC applications
E2-04	N/A	6		6	6	Number of motor poles (Default: 4)
E2-11	N/A	0.20		0.75	1.50	Motor rated output (kW)
H2-01	n057	0				Programs the multi-function relay output MA-MC. (0 = During Run) Note: Do NOT use Default: E
L1-01	n037	N/A		N/A	N/A	Sets motor overload protection to standard fan cooled.
L2-01	-	1				Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L3-01	N/A	0				Stall Prevention accel selection (0=Disabled)
L3-04	n092	0				Stall prevention during decel is disabled.
L3-05	n116	0				Stall Prevention running selection (0=Disabled)
L5-01	-	1				Number of Auto Restart Attempts for L2-01 Condition. Default: 0.
L8-05	N/A	0				Input phase loss protection. (0=Disabled)
L8-35	N/A	1				Side-by-Side Selection (Default: 2)
O3-02	N/A	1				Enables reading of parameter settings. For handheld LCD operator only.
The following parameters are required for multi-speed applications:						
D1-02	n025	50		60 for most applications 55 for Revolution OMS apps	53 for all applications	Sets preset speed 1 (ignores D1-01) when S5 is connected to SC
D1-03	n026	45		50	45	Sets preset speed 2 (ignores D1-01) when S6 is connected to SC
D1-04	n027	40				Sets preset speed 3 (ignores D1-01) when S5 and S6 are connected to SC
The following parameters are required for variable speed applications using a 4-20mA reference signal:						
B5-15	n/a	26 Hz For modulating load cell tray feeders only		Default	Default	PID sleep function start level.
D3-01	n083	9		14	11	Sets jump frequency 1 – Twice this value must be less than the result of H3-12.
		0 (Default) For modulating load cell tray feeders only				

Yaskawa V1000 Inverter Parameter Values for FastBack® Conveyors					
V1000 Parameter Number	V7 Parameter Reference	50E / 90E Values	260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Function Description
D3-04	n086	9 0 (Default) For modulating load cell tray feeders only	14	11	Sets range of jump frequency 1 – Twice this value must be <u>less than</u> H3-12.
H3-02	-	F			Defines the function assigned to Analog input terminal A1. "F" = not used.
H3-04	-	0% (Default)			Terminal A1 bias. Sets the level of the selected input value that is equal to 0 V for A1 input.
H3-09	-	2 (Default)	2 (Default)	2 (Default)	Selects input signal type for Analog input terminal A2. "2" = 4-20 mA.
H3-12	n061	40 50 ** For modulating load cell tray feeders only	47	42	Defines the level of the input value that is equal to 4 mA at terminal A2. Sets the lower run limit of the inverter at H3-12% of E1-04 when given a 4mA analog reference signal. For 90E, 40% $\times$ 50 Hz = 20.0 Hz operation at a 4mA input. For 260E-G2 & G3, 47% $\times$ 60 Hz = 28.2 Hz. For 260E-G3 Plus, 42% $\times$ 53 Hz = 22.3 Hz.
The following parameters are required when using Ethernet communications with the VFD:					
F6-01	-	3			Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3			External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-2 – V1000 Inverter Parameters for FastBack® Conveyors

NOTES:

* These Acceleration / Deceleration values (**C1-01 and C1-02**) are required for maximum operating conditions of 65 Hz speed with maximum-rated incoming power $\leq$ 490 VAC.

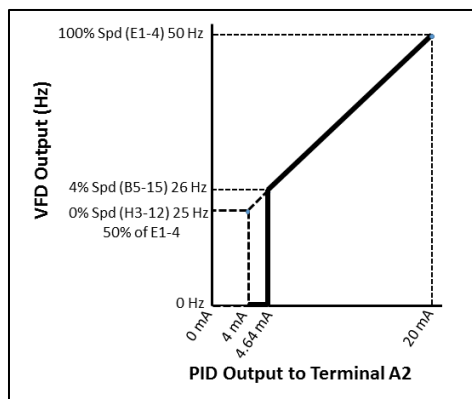


Figure PH009-2 – PID Output to Terminal A2

**** This brings the conveyor to a stop if the PID output gets so low that it is apparent the conveyor should stop, because it is no longer producing travel rate. In practice, H3-12 should be set to the value that produces about 1 ft/min travel rate. Once that is known B5-15 can be set to something just above that.**

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Yaskawa V1000 Inverter Parameter Values for Revolution Equipment

Yaskawa V1000 Inverter Parameter Values for Revolution Equipment						
V1000 Parameter Number	V7 Parameter Reference	Generations 1.0 & 2.0 Revolution Tumble Drum (117RPM /15:1 Gearmotor)	OBSOLETE Generation 1.0 Revolution Gate (117 RPM / 15:1 Gearmotor)	Generation 2.0 Revolution Gate (70 RPM / 25:1 Gearmotor)	Function Description	
The following parameters are required for all applications:						
B1-01	n004	0 (Single / Multi-speed)			Reference freq source selection. For single and multi-speed control: Dipswitch S2 = OFF.	
B1-17	-	1			Run Command at Power Up 1 = Run Command issued, motor operation start	
C1-01	n019	0.50	16	13.10	Sets acceleration time #1 (sec)	These times mate up to D1- 03 (Slow Speed)
C1-02	n020	0.50	0.20	0.20	Sets deceleration time #1 (sec)	
C1-03	n021	N/A	0.50	0.40	Sets acceleration time #2 (sec)	These times mate up to D1- 02 (Fast Speed) when S5 is jumpered to S7.
C1-04	n022	N/A	0.10	0.10	Sets deceleration time #2 (sec)	
C6-02	n080	3 (8.0 kHz) for 208 / 230V			Carrier Frequency selection (default depends on O2-04 for 200V Class)	
		2 (5 kHz) for400 / 460V			Carrier Frequency selection (default depends on O2-04 for 400V Class)	
D1-01	n024	H&C Drum (9 Flights): 80 Hz --- Proprietary and AccuFlavor Drum (16 Flights): 64 Hz	50	60	Sets master speed for single-speed operation (Default: 0 Hz)	
D1-02	n025	N/A	40	67	Sets preset speed 1 (ignores D1-01) when S5 is connected to SC. Fast Speed for Revolution Gate.	
D1-03	n026	N/A	8	13	Sets preset speed 2 (ignores D1-01) when S6 is connected to SC. Slow Speed for Revolution Gate.	
E1-04	n011	80	50	67	Sets max output freq (Default: 60 Hz)	
E1-06	n013	60 (Default)	50	60 (Default)	Sets frequency at which voltage output is maximized.	
E2-01	n036	0.73 (for 208 VAC applications)			Sets motor rated current for 1/8 HP Sumitomo brand gearmotor. Actual nameplate values are 0.66 (230V) and 0.33 (460V).	
		0.66 (for 230 VAC applications)				
		0.40 (for 400 VAC applications)				
		0.33 (for 460 VAC applications)				
E2-03	N/A	0.60 (230V series) / 0.30 (460V series)			Sets motor No-Load Current. Must be <= E2-01.	
E2-11	N/A	0.10	0.10	0.10	Motor rated output (kW)	
H1-07	n056	N/A	7		Programs the multi-function input S7. (7 = Commands to use Accel / Decel time #2 (C1-03 / C1-04). CRITICAL SETTING FOR 2-SPEED OPERATION	

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Yaskawa V1000 Inverter Parameter Values for Revolution Equipment					
V1000 Parameter Number	V7 Parameter Reference	Generations 1.0 & 2.0 Revolution Tumble Drum (117RPM /15:1 Gearmotor)	OBSOLETE Generation 1.0 Revolution Gate (117 RPM / 15:1 Gearmotor)	Generation 2.0 Revolution Gate (70 RPM / 25:1 Gearmotor)	Function Description
H2-01	n057	0	6		Programs the multi-function relay MA-MC output. (0 = During Run, 6 = Drive Ready)
L2-01	-	1			Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	-	1			Number of Auto Restart Attempts for L2-01 Condition. Default: 0.
O3-02	N/A	1			Enables reading of parameter settings. For handheld digital operator only.
The following parameters are required when using Ethernet communications with the VFD:					
F6-01	-	3			Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3			External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-3 – V1000 Inverter Parameters for Revolution Equipment

Yaskawa V1000 Inverter Parameter Values for Stand-Alone Tumble Drum Units

Yaskawa V1000 Inverter Parameter Values for Stand-Alone Tumble Drum Units			
V1000 Parameter Number	V7 Parameter Reference	Values for Stand-Alone Tumble Drum Units	Function Description
<i>The following parameters are required for all applications:</i>			
B1-01	n004	0 (Single Speed)	Reference frequency source selection For single and multi-speed control: Dipswitch S2 = OFF
B1-17	-	1	Run Command at Power Up 1 = Run Command issued, motor operation start
C1-01	n019	0.5	Sets acceleration time 1 in seconds
C1-02	n020	0.5	Sets deceleration time 1 in seconds
D1-01	n024	66 (for 28" dia. Drum w/ SST Flights) 75 (for 32" dia. Drum w/ SST Flights) 80 (for 34" dia. Barrels which accept the 28" and 32" dia. Removable liners) 90 (for 40" dia. Drum w/ SST Flights)	Sets master speed for single-speed operation (Default: 0 Hz)
E1-04	n011	90	Sets maximum output freq (Default: 60 Hz)
E2-01	n036	2.00 (for 230 VAC applications) or 1.00 (for 460 VAC applications)	Sets motor rated current for 1/2 hp, 1800 RPM Baldor motor.
E2-11	N/A	0.40	Motor rated output (kW)
H2-01	n057	0	Programs the multi-function relay output MA-MC. (0 = During Run)
L2-01	-	1	Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	-	1	Number of Auto Restart Attempts for L2-01 Condition. Default : 0.
O3-02	N/A	1	Enables reading of parameter settings. For handheld digital operator only.
<i>The following parameters are required when using Ethernet communications with the VFD:</i>			
F6-01	-	3	Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3	External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-4 – V1000 Inverter Parameters for Tumble Drum Units

Yaskawa V1000 Inverter Parameter Values for Incline / Belted Conveyors

Yaskawa V1000 Inverter Parameter Values for Incline / Belted Conveyors			
V1000 Parameter Number	V7 Parameter Reference	Values for Incline / Belted Conveyors	Function Description
<i>The following parameters are required for all applications:</i>			
B1-01	n004	0 (Single Speed)	Reference frequency source selection For single and multi-speed control: Dipswitch S2 = OFF
B1-17	-	1	Run Command at Power Up 1 = Run Command issued, motor operation start
C1-01	n019	1.00	Sets acceleration time 1 in seconds
C1-02	n020	1.00	Sets deceleration time 1 in seconds
D1-01	n024	60 Note: Vary speed as required for best operation.	Sets master speed for single-speed operation (Default: 0 Hz). If operating higher than 60 Hz, adjust variable E1-04.
E1-03	n0	B	V/f Pattern Selection (Default = F) B = High Starting Torque
E1-04	n011	60 (Default)	Sets maximum output freq (Default: 60 Hz)
E2-01	n036	Varies (230 VAC or 460 VAC applications)	Sets motor rated current. Verify the correct current rating as identified on the motor data plate for the corresponding incoming voltage.
E2-11	N/A	Varies	Motor rated output (kW)
H2-01	n057	0	Programs the multi-function relay output MA-MC. (0 = During Run)
L2-01	-	1	Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	-	1	Number of Auto Restart Attempts for L2-01 Condition. Default: 0.
O3-02	N/A	1	Enables reading of parameter settings. For handheld digital operator only.
<i>The following parameters are required when using Ethernet communications with the VFD:</i>			
F6-01	-	3	Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3	External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-5 – V1000 Inverter Parameters for Incline / Belted Conveyors

Yaskawa V1000 Inverter Parameter Values for Modular Dust Collection Unit

Yaskawa V1000 Inverter Parameter Values for Modular Dust Collection Unit		
V1000 Parameter Number	Values for Modular Dust Collection	Function Description
The following parameters are required for all applications:		
B1-01	0 (Single Speed)	Reference frequency source selection For single and multi-speed control: Dipswitch S2 = OFF
B1-04	1	Reverse disabled. (Default: 0)
B1-17	1	Run Command at Power Up 1 = Run Command issued, motor operation start
C1-01	10.00	Sets acceleration time 1 in seconds
C1-02	3.00	Sets deceleration time 1 in seconds
D1-01	55 50 Note: Vary speed Down as if lower air flow is adequate for the particular dust/powder being used.	Sets master speed for single-speed operation (Default: 0 Hz). 65 Hz is the MAXIMUM rated setting. Lower speeds of 50 – 55 Hz are commonly used.
E1-04	65	Sets maximum output freq (Default: 60 Hz)
E2-01	2.60 (for 200 - 230 VAC applications) 1.30 (for 380 - 460 VAC applications)	Sets motor rated current. Verify the correct current rating as identified on the motor data plate and the incoming voltage.
E2-11	0.75	Motor rated output (kW)
H2-01	0	Programs the multi-function relay output MA-MC. (0 = During Run)
L2-01	1	Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	1	Number of Auto Restart Attempts for L2-01 Condition. Default: 0.
The following parameters are required when using Ethernet communications with the VFD:		
F6-01	3	Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	3	External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-6 – V1000 Inverter Parameters for Modular Dust Collection Unit

Yaskawa V1000 Inverter Parameter Values for AC TecWeigh Seasoning Applicator (S5A)

Yaskawa V1000 Inverter Parameter Values for AC TecWeigh Seasoning Applicator (S5A)				
V1000 Parameter Number	V7 Parameter Numbers	S5A Values		Function Description
The following parameters are required for all applications:				
B1-01	n004	1 (Analog input)		Reference frequency source selection For analog input: Dipswitch S2 = ON. Dipswitch S1 = I (4-20 mA).
B1-04	n006	1		Reverse disabled. (Default: 0)
B1-17	-	1		Run Command at Power Up 1 = Run Command issued, motor operation start
B2-03	-	0.0		Sets DC Injection braking time at start.
B2-04	n090	0.0		Sets DC injection braking time at stop
C1-01	n019	1.20		Sets acceleration time #1 in seconds
C1-02	n020	1.20		Sets deceleration time #1 in seconds
D1-01	n024	60		Sets master speed for single-speed operation (Default: 0 Hz)
D3-01	n083	0 (default)		Sets jump frequency 1 – Twice this value must be <u>less than</u> (below) the result of H3-12.
D3-04	n086	0		Sets range of jump frequency 1 – Twice this value must be <u>less than</u> (below) the result of H3-12.
E1-04	n011	110		Sets maximum output freq for 20 mA input (Default: 60 Hz) Maximum Auger speed at 20 mA = 110 Hz = 247 RPM = 100% Speed
E1-05	n012	230 (208 / 230 VAC applications) or 460 (400 / 460 VAC applications)		Sets maximum voltage out
E1-06	n013	60 (Default)		Sets frequency at which voltage output is maximized.
E1-07	n014	2.00		Mid Output Frequency (Hz)
E1-09	n016	2.00		Sets the minimum allowable running frequency for auger. Minimum Auger speed at 4.29mA = 2.0Hz = 4.5 RPM
E2-01	n036	Black Painted Motor	SST Motor	Sets motor rated current for 1765 RPM Marathon brand motors
		2.00 (208 VAC) or 1.80 (230 VAC) or 1.00 (400 VAC) or 0.90 (460 VAC)	1.60 (208 VAC) or 1.60 (230 VAC) or 0.80 (400 VAC) or 0.80 (460 VAC)	
E2-11	N/A	0.20		Motor rated output (kW)
H2-01	n057	0		Programs the multi-function relay output MA-MC. (0 = During Run)
H3-02	-	F		Defines the function assigned to Analog input terminal A1. "F" = not used.
H3-04	-	0% (Default)		Terminal A1 bias. Sets the level of the selected input value that is equal to 0 V for A1 input.
H3-09	-	2 (Default)		Selects input signal type for Analog input terminal A2. "2" = 4-20 mA.

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Yaskawa V1000 Inverter Parameter Values for AC TecWeigh Seasoning Applicator (S5A)						
V1000 Parameter Number	V7 Parameter Numbers	S5A Values				Function Description
H3-12	n061	0 (Default)				Defines the level of the input value that is equal to 4 mA at terminal A2. Sets the lower run limit of the inverter at H3-12% of E1-04 when given a 4mA analog reference signal. 0% x 110 Hz = 0.0 Hz operation with a 4mA input.
L1-01	n037	3				Sets motor overload protection to vector motor. 100:1 Turndown.
L2-01	-	1				Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	-	1				Number of Auto Restart Attempts for L2-01 Condition. Default: 0
L6-01	n096	4				Enables the Overtorque detection and stops operation (fault oL3)
L6-02	n098	208 VAC 60% = 1.8 amps	230 VAC 54% = 1.6 amps	400 VAC 50% = 0.9 amps	460 VAC 44% = 0.8 amps	Sets the Overtorque detection current level as a percentage of the inverter-rated current (for 200V series = 3.0 amps / for 400V series = 1.8 amps)
L6-03	n099	0.5				Sets Overtorque protection time (seconds). When motor current exceeds the overtorque limit for this time, the drive faults (oL3)
O3-02	N/A	1				Enables reading of parameter settings. For handheld digital operator only.
The following parameters are required when using Ethernet communications with the VFD:						
F6-01	-	3				Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3				External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-7 – V1000 Inverter Parameters for TecWeigh Seasoning Applicator

Yaskawa V1000 Inverter Parameter Values for MinuteMan Accumulator Trolley Conveyors

Yaskawa V1000 Inverter Parameter Values for MinuteMan Accumulator Trolley Conveyors			
V1000 Parameter Number	V7 Parameter Reference	Values for MinuteMan Accumulation Trolley Conveyors	Function Description
The following parameters are required for all applications:			
B1-01	n004	0 (Single Speed)	Reference frequency source selection For single and multi-speed control: Dipswitch S2 = OFF
B1-17	-	1	Run Command at Power Up 1 = Run Command issued, motor operation start
C1-01	n019	1.0	Sets acceleration time 1 in seconds
C1-02	n020	1.0	Sets deceleration time 1 in seconds
D1-01	n024	42 Note: Vary speed as required for best operation.	Sets master speed for single-speed operation (Default: 0 Hz). If operating higher than 60 Hz, adjust variable E1-04 accordingly.
E1-04	n011	60 (Default)	Sets maximum output freq (Default: 60 Hz)
E2-01	n036	1.80 (for 208 VAC applications) or 1.60 (for 230 VAC applications) or 0.80 (for 460 VAC applications)	Sets motor rated current. Verify the correct current rating as identified on the motor data plate for the corresponding incoming voltage.
E2-03	-	1.60 (for 208 VAC applications) or 1.40 (for 230 VAC applications) or 0.70 (for 460 VAC applications)	Sets motor no-load current. Must be < E2-01.
E2-11	N/A	0.37	Motor rated output (kW)
H2-01	n057	0	Programs the multi-function relay output MA-MC. (0 = During Run)
L2-01	-	1	Momentary Power Loss Operation Selection. Default: 0 = Disabled. 1 = Enabled-Recover within L2-02 sec
L5-01	-	1	Number of Auto Restart Attempts for L2-01 Condition. Default: 0.
O3-02	N/A	1	Enables reading of parameter settings. For handheld digital operator only.
The following parameters are required when using Ethernet communications with the VFD:			
F6-01	-	3	Communication Error Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only
F6-03	-	3	External Fault from Communication Option Selection. Default: 1 = Coast to Stop. 3 = Alarm Only

Table PH009-8 – V1000 Inverter Parameters for MinuteMan Accumulation Trolley Conveyors

TECHNICAL PUBLICATION PH009 – YASKAWA V1000 SETUP PROCEDURE

Heat and Control Product Handling Systems

Yaskawa V1000 Inverter Parameter Values for Synchronized SwitchBack™ 260E Conveyors

Yaskawa V1000 Inverter Parameter Values for Synchronized SwitchBack™ 260E Conveyors			
V1000 Parameter Number	Lead Conveyor	Follower Conveyor	Function Description
<i>The following parameters are required for all applications:</i>			
B1-01	0	1 Confirm	Frequency reference. 0: Operator = digital preset speed d1-01 to d1-17 1: Terminals = Analog input terminal A1 or A2.
B1-02	Default: 1 – Confirm	Default: 1 – Confirm	Run command. Default: 1 = Digital input terminals S1 to S7.
B1-04	1	1	Reverse disabled. (Default: 0)
B1-17	1	1	Run command at power up. 1 = Run Command issued, motor operation start
B2-02	3	0	Sets DC injection braking current as a percentage of drive rated current.
B2-03	0.0	0.0	Sets DC Injection braking time at start.
B2-04	0.50 (default)	0.0	Sets DC injection braking time at stop.
B8-01	0 Confirm	0 Confirm	Energy Saving Control Selection 0 = Disabled / 1 = Enabled This feature not applicable to the FastBack mechanism – Do Not Enable
B8-04	447.40 (changes when E2-11 changes)	447.40 (changes when E2-11 changes)	Sets Energy Saving coefficient and fine tunes V/f control.
C1-01	1.00	0.50	Sets acceleration time #1 in seconds.
C1-02	1.50	0.80	Sets deceleration time #1 in seconds.
C4-01	0	0	Torque Compensation Gain (Default: 1.00) CRITICAL SETTING Any value > 0.00 will increase risk of an overvoltage fault
C6-01	0	0	Normal/Heavy Duty Selection (Default: 1)
C6-02	1	1	Carrier Frequency Selection (Default depends on O2-04 for 200V and 400V Class)
D1-01	50		Sets master speed for single-speed operation (Default: 0 Hz)
D1-02	50		Sets preset speed 1 (ignores D1-01) when S5 is connected to SC
D1-03	40		Sets preset speed 2 (ignores D1-01) when S6 is connected to SC
D1-04	35		Sets preset speed 3 (ignores D1-01) when S5 and S6 are connected to SC
E1-04		60 Confirm	Sets max output freq (Default: 60 Hz)
E1-05	130 (208/230V) 260 (400/460V)	130 (208/230V) 260 (400/460V)	Sets maximum voltage out: For 208 / 230 VAC incoming power CRITICAL SETTING For 400 / 460 VAC incoming power
E1-06		60 Confirm	Sets frequency at which voltage output is maximized.
E2-01	4.40 (208V), 4.00 (230V), 2.30 (400V), 2.00 (460V)	4.40 (208V), 4.00 (230V), 2.30 (400V), 2.00 (460V)	Sets motor rated (nameplate) current.
E2-04	6	6	Number of motor poles (Default: 4)
E2-11	0.75	0.75	Motor rated output (kW)
H2-01	0	0	Programs the multi-function relay output MA-MC. (0 = During Run) Note: Do NOT use Default = E
H3-02	F	0 Confirm	Defines the function assigned to Analog input terminal A1. "F" = not used.
H3-10		F	Sets function of terminal A2.